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Refrigerator

Abstract

Disclosed herein is a refrigerator. The refrigerator includes a main body including a storage compartment, a door body rotatably coupled to the main body to open and close the storage compartment, and a front panel detachably coupled to the door body. The door body includes an upper cap including a trim receiving space. The front panel includes an upper trim inserted into the trim receiving space. A voice path is formed in the upper cap and a microphone module is provided in the upper trim to receive voice through the voice path.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) (1) This application is a continuation application, under 35 U.S.C. § 111(a), of international application No. PCT/KR2023/006439, filed on May 11, 2023, which claims priority to Korean Patent Applications No. 10-2022-0111039, filed on Sep. 1, 2022, and No. 10-2022-0136029, filed on Oct. 20, 2022 in the Korean Intellectual Property Office, the disclosure of which is incorporated by reference herein in their entireties

BACKGROUND

1. Field

(1) The disclosure relates to a refrigerator, and more particularly, to a refrigerator including a microphone module configured to receive voice.

2. Description of Related Art

(2) A refrigerator is a home appliance that keeps food fresh by including a main body including a storage compartment, a cold air supply device configured to supply cold air to the storage compartment, and a door configured to open and close the storage compartment.

(3) A microphone module may be provided in the refrigerator to implement voice recognition. Through voice recognition using a microphone module, various functions of the refrigerator may be operated, and Internet of things (IOT) devices connected to the refrigerator may be controlled.

(4) As disclosed in Korean Patent Publication No. 10-2021-0093475, the microphone module may be mounted on a cap-deco that normally forms an upper surface of the door.

SUMMARY

(5) Therefore, it is an aspect of the disclosure to provide a refrigerator including a door body rotatably coupled to a main body to open and close a storage compartment, and a front panel detachably mounted on the door body and including a microphone module arranged on a rear surface thereof.

(6) It is another aspect of the disclosure to provide a refrigerator capable of excluding a microphone hole, which is for receiving voice through a microphone module, from a front panel.

(7) It is another aspect of the disclosure to provide a refrigerator capable of easily connecting a display arranged on a rear surface of a front panel to a microphone module.

(8) Additional aspects of the disclosure will be set forth in part in the description which follows and, in part, will be obvious from the description, or may be learned by practice of the disclosure.

(9) In accordance with an aspect of the disclosure, a refrigerator includes a main body having a storage compartment, and a door body rotatably coupled to the main body to open and close the storage compartment. The door body may include a door outer plate forming a front surface of the door body, and an upper cap coupled to an upper portion of the door outer plate, the upper cap having a fixer opening on an upper side of the upper cap, a trim opening on a front side of the upper door cap, and a trim receiving space. The refrigerator may include a front panel including an upper trim, a fixer, and a microphone module coupled to the upper trim. The door body, the front panel, and the fixer may be configured so that the upper trim is insertable into the trim receiving space through the trim opening, and, with the upper trim inserted into the trim receiving space through the trim opening, the fixer is insertable into the trim receiving space through the fixer

opening to be coupled to the upper trim, to detachably couple the front panel to a front side of the door body with the microphone module disposed in the trim receiving space to receive voice through a voice path formed in the upper cap.

(10) The upper cap may include a cap body, and a front upper wall on a front upper side of the cap body so that the trim receiving space is below the front upper wall. A voice guide hole penetrating the front upper wall may form part of the voice guide path.

(11) The upper cap may include a cap protrusion protruding upward from a periphery of the voice guide hole. A voice guide space may be surrounded by the cap protrusion and form a part of the voice guide path.

(12) The cap protrusion may include a cap front protrusion in front of the voice guide space, and a voice input hole may be formed in the cap front protrusion and may form a part of the voice guide path.

(13) The voice input hole may be spaced apart from a lower surface of the front upper wall by a predetermined height.

(14) The upper cap may include a protrusion upper opening on an upper side of the voice guide space, and the upper cap may include a cap cover that is coupleable to the upper cap to cover the protrusion upper opening.

(15) The cap cover may include a cover body configured to cover the protrusion upper opening, with a lower surface of the cover body forming at least a part of the voice path, and a cover leg extending downward from the lower surface of the cover body and including a cover hooking member configured to hook to a lower surface of the front upper wall.

(16) The upper trim may include a microphone module mounting member protruding rearward, and the microphone module mounting member may include a microphone module mounting space on a lower surface of the microphone module mounting member on which the microphone module is mounted, to thereby be coupled to the upper trim.

(17) The microphone module may include a microphone substrate, and a microphone on a mounting surface of the microphone substrate, and the microphone module may be mounted in the microphone module mounting space with the mounting surface of the microphone substrate to facing downward. The microphone substrate may include a substrate voice hole penetrating the microphone substrate at a position corresponding to the microphone.

(18) The upper trim may include a trim voice hole at a position corresponding to the substrate voice hole.

(19) The refrigerator may include a sealing member between the microphone module mounting member and the microphone module. The sealing member may include a sealing member voice hole at a position corresponding to the trim voice hole and the substrate voice hole.

(20) The upper trim may include a trim protrusion protruding upward and configured to be coupled to the cap cover in a state in which the upper trim is inserted into the trim receiving space and the cap cover is coupled to the upper cap.

(21) The microphone module may include a connector electrically connected to an external circuit, and the connector may be on the mounting surface of the microphone substrate and positioned below the microphone substrate.

(22) The upper trim may include a connector receiving member configured to receive the connector or other connector connected to the connector.

(23) The microphone module may include a microphone substrate, and a plurality of microphones spaced apart from each other on a mounting surface of the microphone substrate. The voice path may include a plurality of voice paths separated from each other to correspond to the plurality of microphones.

(24) In accordance with another aspect of the disclosure, a refrigerator includes a main body comprising a storage compartment; a door body rotatably coupled to the main body to open and close the storage compartment, the door body comprising a door outer plate forming a front surface

of the door body and an upper cap coupled to an upper portion of the door outer plate and comprising a trim receiving space; a front panel detachably coupled to a front side of the door body and comprising an upper trim inserted into the trim receiving space through a trim opening formed on a front side of the upper cap; a fixer inserted into the trim receiving space through a fixer opening formed on an upper side of the upper cap, the fixer coupled to the upper trim; and a microphone module coupled to the upper trim and disposed in the trim receiving space together with the upper trim, wherein the upper cap comprises: a cap body; a front upper wall arranged on a front upper side of the cap body, and provided in such a way that the trim receiving space is formed below the front upper wall; and a stepped member formed by bending upward from the front upper wall to allow a voice path provided to receive voice to be formed between the front panel and the stepped member.

(25) In accordance with another aspect of the disclosure, a home appliance includes a main body providing a cavity, a door body rotatably coupled to the main body to open and close the cavity, and having a trim receiving space in an upper portion of the door body, a front panel detachably coupled to a front side of the door body and including an upper trim protruding rearward from an upper portion of a rear surface of the front panel and received in the trim receiving space, and a microphone module coupled to the upper trim and disposed in the trim receiving space together with the upper trim. The door body may have a voice path between an outside of the door body and the trim receiving space to allow voice to be received by the microphone module.

(26) The home appliance may further include a display on the rear surface of the front panel and configured to display an image through the front panel, and a display substrate provided on the rear surface of the front panel to drive the display.

(27) The microphone module and the display substrate may be electrically connected to each other through a wire.

(28) The upper trim may include a microphone module mounting member protruding rearward. The microphone module mounting member may include a microphone module mounting space on a lower surface of the microphone module mounting member, and on which the microphone module is mounted. The upper trim may have a trim voice hole penetrating the upper trim to guide voice to the microphone module.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) These and/or other aspects of the disclosure will become apparent and more readily appreciated from the following description of embodiments, taken in conjunction with the accompanying drawings of which:

(2) FIG. 1 is a view illustrating a front appearance of a refrigerator according to an embodiment of the disclosure;

(3) FIG. 2 is a view illustrating a state in which a door of the refrigerator is opened according to an embodiment of the disclosure;

(4) FIG. 3 is a view illustrating an upper portion of the door according to an embodiment of the disclosure;

(5) FIG. 4 is a view illustrating a state in which a door body and a front panel of the door are separated according to an embodiment of the disclosure;

(6) FIG. 5 is an exploded view illustrating the door body according to an embodiment of the disclosure;

(7) FIG. 6 is a view illustrating a rear surface of the front panel according to an embodiment of the disclosure;

(8) FIG. 7 is an exploded view illustrating the front panel according to an embodiment of the

disclosure;

- (9) FIG. 8 is a view illustrating an operation of coupling the front panel to the door body according to an embodiment of the disclosure, particularly illustrating an operation of coupling a lower end of the front panel to a lower cap;
- (10) FIG. 9 is a view illustrating an operation of coupling the front panel to the door body according to an embodiment of the disclosure, particularly illustrating a state in which the front panel is rotated around the lower cap following the operation of FIG. 8;
- (11) FIG. 10 is a view illustrating an operation of coupling the front panel to the door body according to an embodiment of the disclosure, particularly illustrating a state in which a fixer is coupled to an upper end of the front panel and an upper cap following the operation of FIG. 9;
- (12) FIG. 11 is a view illustrating an upper fixer and a microphone module according to an embodiment of the disclosure;
- (13) FIG. 12 is a bottom view of the upper fixer and the microphone module according to an embodiment of the disclosure, particularly illustrating a state in which the upper fixer and the microphone module are separated;
- (14) FIG. 13 is a bottom view of the upper fixer and the microphone module according to an embodiment of the disclosure, particularly illustrating a state in which the microphone module is mounted to the upper fixer;
- (15) FIG. 14 is an enlarged view of an upper portion of the upper cap according to an embodiment of the disclosure;
- (16) FIG. 15 is a view illustrating a state in which a cap cover is separated from the upper cap according to an embodiment of the disclosure;
- (17) FIG. 16 is a perspective cross-sectional view illustrating the microphone module attached to an upper trim, and a voice path formed in the upper cap according to an embodiment of the disclosure;
- (18) FIG. 17 is a view illustrating the microphone module attached to the upper trim and the voice path formed in the upper cap according to an embodiment of the disclosure, particularly a cross-sectional view taken along a Y-Z plane;
- (19) FIG. 18 is a view illustrating the microphone module attached to the upper trim and the voice path formed in the upper cap according to an embodiment of the disclosure, particularly a cross-sectional view taken along a X-Y plane;
- (20) FIG. 19 is other view illustrating the microphone module attached to the upper trim and the voice path formed in the upper cap according to an embodiment of the disclosure, particularly a cross-sectional view taken along the X-Y plane; and
- (21) FIG. 20 is a perspective cross-sectional view illustrating a microphone module attached to an upper trim, and a voice path formed between a stepped member of an upper cap and a front panel according to an embodiment of the disclosure.

DETAILED DESCRIPTION

- (22) The various embodiments and the terms used therein are not intended to limit the technology disclosed herein to specific forms, and the disclosure should be understood to include various modifications, equivalents, and/or alternatives to the corresponding embodiments.
- (23) In describing the drawings, similar reference numerals may be used to designate similar constituent elements.
- (24) A singular expression may include a plural expression unless they are definitely different in a context.
- (25) The expressions “A or B,” “at least one of A or/and B,” or “one or more of A or/and B,” A, B or C,” “at least one of A, B or/and C,” or “one or more of A, B or/and C,” and the like used herein may include any and all combinations of one or more of the associated listed items.
- (26) The term of “and/or” includes a plurality of combinations of relevant items or any one item among a plurality of relevant items.

(27) Herein, the expressions “a first”, “a second”, “the first”, “the second”, etc., may simply be used to distinguish an element from other elements, but is not limited to another aspect (e.g., importance or order) of elements.

(28) When an element (e.g., a first element) is referred to as being “(functionally or communicatively) coupled,” or “connected” to another element (e.g., a second element), the first element may be connected to the second element, directly (e.g., wired), wirelessly, or through a third element.

(29) In this disclosure, the terms “including”, “including”, and the like are used to specify features, numbers, steps, operations, elements, elements, or combinations thereof, but do not preclude the presence or addition of one or more of the features, elements, steps, operations, elements, elements, or combinations thereof.

(30) When an element is said to be “connected”, “coupled”, “supported” or “contacted” with another element, this includes not only when elements are directly connected, coupled, supported or contacted, but also when elements are indirectly connected, coupled, supported or contacted through a third element.

(31) Throughout the description, when an element is “on” another element, this includes not only when the element is in contact with the other element, but also when there is another element between the two elements.

(32) A refrigerator according to an embodiment may include a main body.

(33) The “main body” may include an inner case and an outer case disposed outside the inner case, and an insulating material provided between the inner case and the outer case.

(34) The “inner case” may include a case, a plate, a panel or a liner that forms a storage compartment. The inner case may be formed as a single body or may be formed by assembling a plurality of plates. The “outer case” may form an exterior of the main body, and may be coupled to the outer side of the inner case to allow an insulating material to be disposed between the inner case and the outer case.

(35) The “insulating material” may insulate the inside and outside of the storage compartment to allow a temperature inside the storage compartment to be maintained at a set appropriate temperature without being affected by the external environment of the storage compartment. According to an embodiment, the insulating material may include a foam insulating material. After fixing the inner and outer cases with a jig or the like, a foam insulating material may be molded by injecting and foaming urethane foam, in which polyurethane and a foaming agent are mixed, between the inner and outer cases.

(36) According to an embodiment, the insulating material may include a vacuum insulating material in addition to the foam insulating material, or the insulating material may include only a vacuum insulating material instead of the foam insulating material. The vacuum insulating material may include a core material and an outer covering material that accommodates the core material and seals an inside with a vacuum or a pressure close to the vacuum. The vacuum insulating material may further include an adsorbent that adsorbs gas and moisture to stably maintain a vacuum state.

(37) However, the insulating material is not limited to the above-described foam insulating material or vacuum insulating material, and may include various materials that may be used for insulation.

(38) The refrigerator according to an embodiment may include a storage compartment provided inside the main body to store food.

(39) The “storage compartment” may include a space defined by the inner case. The storage compartment may further include an inner case provided to define the space. The storage compartment may be formed to allow at least one side thereof to be opened in order to take in and out of food. The storage compartment may be provided to store “food”. Food includes food that may be eaten or drunk, and specifically, may include meat, fish, seafood, fruits, vegetables, water, ice, beverages, kimchi or alcoholic beverages such as wine. However, in addition to food,

medicines or cosmetics may be stored in the storage compartment, and there is no limitation in the items that may be stored in the storage compartment.

(40) The refrigerator may include one or more storage compartments. When two or more storage compartments are formed in the refrigerator, each storage compartment may have a different purpose and may be maintained at different temperatures. For this, each of the storage compartments may be partitioned from each other by a partition wall including the insulating material. According to an embodiment, the partition wall may be a part of the main body. According to an embodiment, the partition wall may be a separate partition provided separately from the main body, and assembled to the main body.

(41) The storage compartment may be provided to be maintained in an appropriate temperature range according to the purpose, and may include a “refrigerating compartment”, a “freezing compartment” or a “variable temperature compartment” which are classified according to the purpose and/or temperature range. The refrigerating compartment may be maintained at a temperature suitable for storing food in a refrigerating manner, and the freezing compartment may be maintained at a temperature suitable for storing food in a freezing manner. “Refrigerating” may mean to cool food to the extent that it does not freeze, and as an example, the refrigerating compartment may be maintained in the range of 0 degrees Celsius to 7 degrees Celsius. “Freezing” may mean to freeze or cool food to remain frozen, and as an example, the freezing compartment may be maintained in the range of minus 20 degrees Celsius to minus 1 degree Celsius. The variable temperature compartment may be used as either the refrigerating compartment or the freezing compartment according to a user's choice or regardless of a user's choice. According to an embodiment, a portion of one storage compartment may be used as the refrigerating compartment and the remaining portion may be used as the freezing compartment.

(42) The storage compartment may be called “vegetable compartment”, “fresh compartment”, “cooling compartment” and “ice making compartment” in addition to names such as “refrigerating compartment”, “freezing compartment” and “variable temperature compartment”. Terms such as “refrigerating compartment”, “freezing compartment” and “variable temperature compartment” should be understood as encompassing storage compartments including corresponding uses and temperature ranges, respectively.

(43) The refrigerator according to an embodiment may include a door configured to open and close one open side of the storage compartment.

(44) The “door” may be configured to seal the storage compartment when the door is closed. In the similar manner as the main body, the door may include an insulating material to insulate the storage compartment when the door is closed.

(45) The refrigerator according to an embodiment may include a cold air supply device configured to supply cold air to the storage compartment.

(46) The “cold air supply device” may include a machine, an appliance, an electronic device, and/or a system corresponding to a combination thereof, which are capable of generating and guiding cold air to cool the storage compartment.

(47) According to an embodiment, the cold air supply device may generate cold air through a refrigeration cycle including compression, condensation, expansion, and evaporation of a refrigerant. To this, the cold air supply device may include a refrigeration cycle device including a compressor, a condenser, an expansion device, and an evaporator capable of driving the refrigeration cycle. According to an embodiment, the cold air supply device may include a semiconductor such as a thermoelectric element. The thermoelectric element may cool the storage compartment by heating and cooling through the Peltier effect.

(48) The refrigerator according to an embodiment may include a machine room in which at least some components belonging to the cold air supply device are disposed.

(49) The “machine room” may be provided to be partitioned and insulated from the storage compartment in order to prevent heat, which is generated from components disposed in the

machine room, from being transferred to the storage compartment. The inside of the machine room may be configured to communicate with the outside of the main body so as to dissipate heat from components disposed inside the machine room.

(50) The refrigerator according to an embodiment may include a controller for controlling the refrigerator.

(51) The “controller” may include a memory for storing or memorizing a program and/or data for controlling the refrigerator, and a processor for outputting a control signal for controlling the cold air supply device according to the program and/or data stored in the memory.

(52) The memory may memorize programs and/or data for controlling components included in the refrigerator, and memorize temporary data that is generated while generating control signals for controlling components included in the refrigerator.

(53) The processor may process a user input of the user interface according to a program and/or data memorized/stored in the memory, and may generate a control signal for controlling an operation of the user interface and/or the cold air supply device. For example, the processor may receive temperature information of the storage compartment from a temperature sensor and generate a cooling control signal for controlling the operation of the cold air supply device based on the temperature information of the storage compartment. The processor may receive a user input from the user interface, and transmit a display control signal and image data, which is for displaying an image on the user interface, to the user interface in response to the user input.

(54) The controller may include the processor and the memory configured to control all components included in the refrigerator, and may further include a plurality of processors and a plurality of memories configured to individually control components of the refrigerator. For example, the controller may separately include a processor and a memory configured to control the operation of the cold air supply device according to the output of the temperature sensor, and a processor and a memory configured to control the operation of the user interface according to the user input. The processor and the memory may be provided integrally or separately. The processor may include a plurality of processors, and the memory may include a plurality of memories.

(55) Reference will now be made in detail to embodiments of the disclosure, examples of which are illustrated in the accompanying drawings. Hereinafter the X-axis, Y-axis, and Z-axis are based on the coordinate axes shown in FIG. 2.

(56) FIG. 1 is a view illustrating a front appearance of a refrigerator according to an embodiment of the disclosure. FIG. 2 is a view illustrating a state in which a door of the refrigerator is opened according to an embodiment of the disclosure.

(57) Referring to FIGS. 1 and 2, a refrigerator 1 according to an embodiment may include a main body 10, storage compartments 21, 22, and 23 formed inside the main body 10, doors 31, 32, 33, and 34 configured to open and close the storage compartments 21, 22, and 23, and a cold air supply device (not shown) configured to supply cold air to the storage compartments 21, 22, and 23.

(58) The main body 10 may include an inner case 11 forming the storage compartments 21, 22, and 23, an outer case 12 coupled to the outside of the inner case 11 to form an exterior, and an insulating material (not shown) provided between the inner case 11 and the outer case 12 to insulate the storage compartments 21, 22, and 23.

(59) The storage compartments 21, 22, and 23 may be divided into a plurality of compartments by a horizontal partition wall 24 and a vertical partition wall 25. The storage compartments 21, 22, and 23 may be partitioned into an upper storage compartment 21 and lower storage compartments 22, and 23 by the horizontal partition wall 24, and the lower storage compartments 22 and 23 may be partitioned into a right lower storage compartment 22 and a left lower storage compartment 23 by the vertical partition wall 25.

(60) The upper storage compartment 21 may be used as a refrigerating compartment, and the lower storage compartments 22 and 23 may be used as a freezing compartment. However, the division and use of the storage compartments 21, 22, and 23 as described above is only an example, and is

not limited thereto.

(61) A shelf **26** on which foods are placed, and a storage container **27** storing food may be provided inside the storage compartments **21**, **22**, and **23**.

(62) The cold air supply device may generate cold air using a refrigeration circulation cycle of compressing, condensing, expanding, and evaporating a refrigerant, and supply the generated cold air to the storage compartments **21**, **22**, and **23**.

(63) The storage compartment **21** may be opened and closed by a pair of doors **31** and **32**. The doors **31** and **32** may be rotatably coupled to the main body **10**. The storage compartment **22** may be opened and closed by a door **33**, and the door **33** may be rotatably coupled to the main body **10**. The storage compartment **23** may be opened and closed by a door **34**, and the door **34** may be rotatably coupled to the main body **10**. Hinges **35**, **36**, and **37** may be provided in the main body **10** to rotatably couple the doors **31**, **32**, **33**, and **34** to the main body **10**.

(64) A door shelf **38** on which foods are placed, and a gasket **39** provided to, when the doors **31**, **32**, **33** and **34** are closed, be in close contact with a front surface of the main body **10** may be provided on a rear surface of the doors **31**, **32**, **33** and **34**.

(65) A display **180** may be provided on the doors **31**, **32**, **33** and **34**. However, the display **180** is not essential, and the display **180** may not be provided according to embodiments. The display **180** may display various information related to the state or operation of the refrigerator or display various applications for user convenience. In the drawings of the disclosure, the display **180** is provided on the door **32** on the upper left side, but the display **180** may be provided on other doors **31**, **33**, and **34**. Hereinafter the door **32** on the upper left side will be described, but the following description may be equally applied to the other doors **31**, **33** and **34**.

(66) FIG. **3** is a view illustrating an upper portion of the door according to an embodiment of the disclosure. FIG. **4** is a view illustrating a state in which a door body and a front panel of the door separated according to an embodiment of the disclosure. FIG. **5** is an exploded view illustrating the door body according to an embodiment of the disclosure. FIG. **6** is a view illustrating a rear surface of the front panel according to an embodiment of the disclosure. FIG. **7** is an exploded view illustrating the front panel according to an embodiment of the disclosure. FIG. **8** is a view illustrating an operation of coupling the front panel to the door body according to an embodiment of the disclosure, particularly illustrating an operation of coupling a lower end of the front panel to a lower cap. FIG. **9** is a view illustrating an operation of coupling the front panel to the door body according to an embodiment of the disclosure, particularly illustrating a state in which the front panel is rotated around the lower cap following the operation of FIG. **8**. FIG. **10** is a view illustrating an operation of coupling the front panel to the door body according to an embodiment of the disclosure, particularly illustrating a state in which a fixer is coupled to an upper end of the front panel and an upper cap following the operation of FIG. **9**.

(67) Referring to FIGS. **3** to **10**, the door **32** according to an embodiment of the disclosure may include a door body **40** and a front panel **90** detachably coupled to a front surface of the door body **40**.

(68) The door body **40** may be rotatably coupled to the main body **10** to open and close the storage compartment **21**. The door body **40** may include a door outer plate **41**, a door inner plate **55**, an upper cap **60**, and a lower cap **80** (refer to FIG. **5**). The outer door plate **41**, the inner plate **55**, the upper cap **60**, and the lower cap **80** may be coupled to each other to form an inner space. An insulating material **56** (refer to FIG. **8**) may be provided in the inner space of the door body **40**.

(69) The door outer plate **41** may include a main plate **42** including a main plate opening **44**, and a first heat dissipation plate **51** and a second heat dissipation plate **52** coupled to the main plate opening **44** to dissipate heat emitted from the display **180**.

(70) The main plate **42** may include a front member **43** forming the front surface of the door body **40**. The main plate **42** may include side members **46** forming both left and right sides of the door body **40**. The main plate **42** may be formed in a bent shape to include the front member **43** and the

both side members **46**. However, the front member **43** and the side member **46** may be formed separately and assembled with each other.

(71) A holder mounting groove **45**, into which a holder **47** for coupling with the front panel **90** is mounted, may be formed on the left and right edges of the front member **43**.

(72) The main plate opening **44** may be formed in the center of the front member **43**. The heat dissipation plates **51** and **52** may form a display accommodating member **54** (refer to FIG. 4) concavely formed to accommodate the display **180**. The heat dissipation plates **51** and **52** may include the first heat dissipation plate **51**, and the second heat dissipation plate **52** coupled to an outer surface of the first heat dissipation plate **51**.

(73) The second heat dissipation plate **52** may include a heat dissipation plate opening **53** formed at a central portion of the second heat dissipation plate **52**. The first heat dissipation plate **51** may be coupled to the heat dissipation plate opening **53** of the second heat dissipation plate **52**.

(74) The first heat dissipation plate **51** may be formed of a material having high thermal conductivity to efficiently dissipate heat emitted from the display **180** to the surroundings. For example, the first heat dissipation plate **51** may be formed of a metal material. The second heat dissipation plate **52** may be injection molded from a plastic material to easily mold a structure for coupling with the main plate **42** or a structure for guiding cables.

(75) As described above, the outer door plate **41** may include the first heat dissipation plate **51** and the second heat dissipation plate **52** to efficiently dissipate heat emitted from the display **180**.

Alternatively, the first heat dissipation plate **51** may be directly coupled to the main plate opening **44** without the second heat dissipation plate **52**. Further, the outer door plate **41** may be composed of only the main plate **42** without the first heat dissipation plate **51** and the second heat dissipation plate **52**. In this case, the main plate opening **44** of the main plate **42** may be omitted.

(76) The upper cap **60** and the lower cap **80** may be injection molded from a plastic material. A structure that allows the front panel **90** to be detachably mounted to the door body **40** may be formed on the upper cap **60** and the lower cap **80**. This will be described later. A handle provided to be gripped to open and close the door **32** may be recessed on a lower surface of the lower cap **80**. However, in the lower doors **33** and **34**, the handle may be formed on the upper cap instead of the lower cap.

(77) The front panel **90** may be detachably mounted on the front surface of the door body **40**. The front panel **90** may be provided to cover the entire front region of the door body **40**. The front panel **90** may be formed to have a size corresponding to the size of the front surface of the door body **40**.

(78) The front panel **90** may include a panel body **91**, a lower trim **130** provided on a lower end of a rear surface of the panel body **91**, and an upper trim **150** provided on an upper end of the rear surface of the panel body **91** (refer to FIGS. 6 and 7). The lower trim **130** and the upper trim **150** may be formed separately from the panel body **91** and attached to the panel body **91**. Alternatively, the lower trim **130** and the upper trim **150** may be integrally formed with the panel body **91**.

(79) The panel body **91** may be formed of a glass material. However, it is not limited thereto, and the panel body **91** may be formed of a steel plate material or a plastic material.

(80) The panel body **91** may include a transparent area TA which is seen through and a non-transparent area NA which is not seen through. The transparent area TA may be provided in the central portion of the panel body **91**. The inside of the panel body **91** may be seen through the transparent area TA from the outside of the refrigerator.

(81) The panel body **91** may include a glass member **92** that is seen through, and a decorative layer **93** attached to a rear surface of the glass member **92** (refer to FIGS. 9 and 10). The glass member **92** may include glass or plastic that is transparent the same as glass. The decorative layer **93** may be formed of an opaque material which is not seen through. The decorative layer **93** may be formed on a portion of the rear surface of the glass member **92**.

(82) As mentioned above, by the decorative layer **93** formed on the rear surface of the glass member **92**, the panel body **91** may include the transparent area TA and the non-transparent area

NA. The transparent area TA may be an area in which the decorative layer **93** is not formed. The non-transparent area NA may be an area in which the decorative layer **93** is formed.

(83) A screen of the display **180** disposed behind the panel body **91** may be exposed through the transparent area TA. The color and pattern of the decorative layer **93** may be seen from the front outsider side of the refrigerator through the non-transparent area NA.

(84) The front panel **90** may further include a shatterproof film **94** provided to prevent the glass member **92** from scattering when the glass member **92** is broken. The shatterproof film **94** may be provided on the rear surface of the decorative layer **93**.

(85) The front panel **90** may further include a touch film (not shown) for inputting a signal to the display **180**. The touch film may be provided inside the transparent area TA on the rear surface of the glass member **92** to correspond to the screen of the display **180**.

(86) The front panel **90** may include the display **180** provided on the rear surface of the panel body **91**. The display **180** may be provided on the rear surface of the front panel **90** in such a way that a screen, on which an image is displayed, faces forward. The display **180** may be provided inside the transparent area TA on the rear surface of the panel body **91**. Accordingly, the screen of the display **180** may be displayed to the outside through the transparent area TA. The screen of the display **180** may substantially correspond to the size and shape of the transparent area TA.

(87) The front panel **90** may include a support plate **100**. The support plate **100** may fix the display **180** to the rear surface of the front panel **90** to allow the screen of the display **180** to face forward. The support plate **100** may be coupled to the rear surface of the panel body **91**. The display **180** may be mounted on the rear side of the support plate **100**.

(88) The support plate **100** may be fixed to the rear surface of the panel body **91** through an adhesive such as silicon or hot melt. The support plate **100** may include a display mounting member **101** on which the display **180** is mounted. The support plate **100** may include a substrate mounting member **110** on which a display substrate **190** configured to drive the display **180** is mounted. The substrate mounting member **110** may be formed in a box shape with an open rear surface. The substrate mounting member **110** may be formed below the display mounting member **101**.

(89) Accordingly, because the display substrate **190** is arranged vertically without overlapping the display **180**, a thickness of the front panel **90** may be reduced. The front panel **90** may include a substrate cover **191** to cover the display substrate **190** mounted on the substrate mounting member **110**. As described above, according to an embodiment of the disclosure, the display **180** and the display substrate **190** may be mounted on the rear surface of the panel body **91** and integrated with the panel body **91**. Accordingly, it is easy to connect the display **180** and the display substrate **190** with a wire.

(90) Alternatively, the display **180** may be mounted on the panel body **91**, but the display substrate **190** may be mounted on the door body **40** instead of the panel body **91**.

(91) After the support plate **100** is fixed to the rear surface of the panel body **91**, the display **180** may be mounted on the display mounting member **101** of the support plate **100**. The display **180** may be mounted on the rear side of the display mounting member **101**. The display **180** may be mounted on the display mounting member **101** in a direction from rear to front. Due to the above structure, when removing or replacing the display **180**, there is no need to separate the support plate **100** from the panel body **91**. Accordingly, even in a state in which the support plate **100** is mounted on the panel body **91**, it is possible to separate the display **180** from the support plate **100** and replace the display **180**.

(92) The support plate **100** may include a support plate opening **102** having a size corresponding to the screen of the display **180** to allow the screen of the display **180** not to be covered by the support plate **100**.

(93) The display mounting member **101** may have a rectangular frame shape. The display mounting member **101** may support a front edge of the display **180**. The display mounting member **101** may

support an upper surface, a lower surface, a left surface and a right surface of the display **180**.

(94) The support plate **100** may include a guide rib **111** provided to guide the display **180** to the display mounting member **101**.

(95) The support plate **100** may include a locking hook **113** provided to be elastically deformable, so as to fix the display **180** mounted on the display mounting member **101**. The locking hook **113** may be elastically deformed to open outward while the display **180** is mounted on the display mounting member **101**, and when the display **180** is completely mounted on the display mounting member **101**, the locking hook **113** may be restored to its original state so as to support the display **180**.

(96) After the display **180** is inserted into the display mounting member **101**, the display **180** may be fixed to the display mounting member **101** by a fastening member **116**.

(97) The support plate **100** may include a cable guide **107** provided to guide a first wire cable **117** and a second wire cable **119**. The cable guide **107** may include a rounded member of the substrate mounting member **110**.

(98) The first wire cable **117** may electrically connect the display substrate **190** and a power supply device (not shown) provided in the main body **10**. The display substrate **190** may receive power through the first wire cable **117**. The second wire cable **119** may electrically connect the display substrate **190** to a microphone module **170**. The display substrate **190** and the microphone module **170** may exchange power and signals through the second wire cable **119**.

(99) A first connector **118** may be provided at an end, which is opposite to the display substrate **190**, of the first wire cable **117**. A second connector **120** may be provided at an end, which is opposite to the display substrate **190**, of the second wire cable **119**. When the front panel **90** is coupled to the door body **40**, the first connector **118** may be coupled to a third connector **58** provided on the side of the door body **40**. Through a third wire cable **57**, the third connector **58** may be electrically connected to the power supply provided in the main body **10**. The third wire cable **57** may connect the third connector **58** to the power supply provided in the main body **10** through a hinge shaft provided to connect the door body **40** and the main body **10**. The second connector **120** may be coupled to a microphone connector **174** (refer to FIG. **12**) of the microphone module **170**.

(100) The front panel **90** may include the lower trim **130** and the upper trim **150** for coupling with the door body **40**. The lower trim **130** may be provided at the lower end of the rear surface of the panel body **91**. The lower trim **130** may be formed separately from the panel body **91** and attached to the panel body **91**. The lower trim **130** may be attached to the panel body **91** using an adhesive **132**. Alternatively, the lower trim **130** may be integrally formed with the panel body **91**. The lower trim **130** may be elongated in the left and right direction (X-axis direction). The lower trim **130** may be elongated along the lower edge of the panel body **91**.

(101) When the front panel **90** is coupled to the door body **40**, the lower trim **130** may be supported by the lower cap **80**. When the front panel **90** is coupled to the door body **40**, the lower surface of the lower trim **130** may be covered by the lower cap **80**. Accordingly, the lower trim **130** may not be exposed to the outside.

(102) The lower trim **130** may include a lower trim groove **131**. The lower trim groove **131** may open downward. However, according to embodiments, the lower trim **130** may include a through hole formed to vertically penetrate the lower trim **130** instead of a downwardly open groove. When the front panel **90** is coupled to the door body **40**, a lower cap protrusion **83** of the lower cap **80** may be inserted into the lower trim groove **131**.

(103) In a state in which the front panel **90** is inclined, the lower cap protrusion **83** may be inserted into the lower trim groove **131**. In a state in which the lower cap protrusion **83** is inserted into the lower trim groove **131**, the front panel **90** may rotate around the lower cap protrusion **83**. That is, the front panel **90** may move from an inclined state to an upright state by rotating around the lower cap protrusion **83** in a state in which the lower cap protrusion **83** is inserted into the lower trim

groove **131**. Alternatively, the front panel **90** may move from the upright state to the inclined state in the same way.

(104) Referring to FIG. **12**, the upper trim **150** may be provided at an upper end of the rear surface of the panel body **91**. The upper trim **150** may be formed separately from the panel body **91** and attached to the panel body **91**. The upper trim **150** may be attached to the rear surface of the panel body **91** using an adhesive **161**. Alternatively, the upper trim **150** may be integrally formed with the panel body **91**. The upper trim **150** may be elongated in the left and right direction (X-axis direction). The upper trim **150** may be elongated along the upper edge of the panel body **91**.

(105) When the front panel **90** is coupled to the door body **40**, the upper trim **150** may be covered by the upper cap **60**. Accordingly, the upper trim **150** may not be exposed to the outside.

(106) The upper trim **150** may include an upper trim body **151** attached to the rear surface of the panel body **91** and having a flat shape. The upper trim body **151** may include an adhesive surface on which the upper trim **150** is attached to the panel body **91**.

(107) The upper trim **150** may include a fixer coupler **152** protruding rearward (−Z direction). The fixer coupler **152** may extend rearward from the upper trim body **151**. The fixer coupler **152** and the upper trim body **151** may be integrally formed. The fixer coupler **152** may extend rearward from the upper portion of the upper trim body **151**. The fixer coupler **152** may be provided at both ends of the upper trim **150** in the left and right direction (refer to FIG. **11**).

(108) The fixer coupler **152** may be inserted into the upper cap **60**. For example, the fixer coupler **152** may be inserted into a trim receiving space **63** of the upper cap **60**. A trim opening **64** may be formed on the front side of the trim receiving space **63** of the upper cap **60**. The fixer coupler **152** of the upper trim **150** may be inserted into the trim receiving space **63** of the upper cap **60** through the trim opening **64** of the upper cap **60**.

(109) The fixer coupler **152** may include a fixer coupling hole **153**. The fixer coupling hole **153** may be formed to vertically penetrate the fixer coupler **152**. A fixer **79** may be coupled to the upper trim **150**. For example, the fixer **79** may be coupled to the fixer coupling hole **153** of the upper trim **150**. The fixer **79** may be inserted into the fixer coupling hole **153**. A fixer opening **65** may be formed on the upper side of the trim receiving space **63** of the upper cap **60**. At least one portion of the fixer **79** may be inserted into the trim receiving space **63** of the upper cap **60** through the fixer opening **65** of the upper cap **60**.

(110) When the fixer coupler **152** of the upper trim **150** is inserted into the trim receiving space **63** of the upper cap **60**, the fixer **79** may be inserted in the trim receiving space **63** of the upper cap **60** through the fixer opening **65** of the upper cap **60**, and the fixer **79** may be coupled to the fixer coupler **152** of the upper trim **150**. Due to the above configuration, the front panel **90** may be coupled to the door body **40**.

(111) The upper trim **150** may include a microphone module mounting member **154** (refer to FIGS. **11** to **13**) protruding rearward (−Z direction). The microphone module **170** may be mounted on the microphone module mounting member **154**. The microphone module mounting member **154** may extend rearward from the upper trim body **151**. The microphone module mounting member **154** and the upper trim body **151** may be integrally formed. The microphone module mounting member **154** and the fixer coupler **152** may be integrally formed.

(112) The microphone module mounting member **154** may extend rearward from the upper portion of the upper trim body **151**. The microphone module mounting member **154** may be provided at the center side of the upper trim **150** with respect to the left and right direction. The fixer coupler **152** may be provided on both sides of the microphone module mounting member **154**.

(113) The microphone module mounting member **154** may be inserted into the upper cap **60**. For example, the microphone module mounting member **154** may be inserted into the trim receiving space **63** of the upper cap **60**. The trim opening **64** may be formed on the front side of the trim receiving space **63** of the upper cap **60**. The microphone module mounting member **154** of the upper trim **150** may be inserted into the trim receiving space **63** of the upper cap **60** through the

trim opening **64** of the upper cap **60**.

(114) The front panel **90** may include a side trim **140** for coupling with the door body **40**. The side trim **140** may be provided to extend vertically on the left and right edge of the panel body **91**. The side trim **140** may be formed separately from the panel body **91** and attached to the panel body **91**. Alternatively, the side trim **140** may be integrally formed with the panel body **91**.

(115) The side trim **140** may be coupled to the holder **47** mounted on the door body **40**. As the side trim **140** is coupled to the holder **47**, both sides of the front panel **90** may be in close contact with the door body **40**.

(116) According to embodiments, the front panel **90** may include a magnet instead of the side trim **140**. In this case, the holder mounting groove **45** and the holder **47** of the door body **40** may be omitted. The front member **43** of the door body **40** may be formed of an iron plate material so as to be in close contact with the magnet.

(117) Referring to FIGS. **8** and **9**, the lower cap **80** of the door body **40** may be coupled to the lower portion of the door outer plate **41**. The lower cap **80** may form the lower surface of the door body **40**. The lower cap **80** may include a lower cap body **81**. The lower cap **80** may include a lower cap support **82** protruding forward from a lower portion of the lower cap body **81**. The lower cap support **82** may be formed to protrude forward from the door outer plate **41**. The lower cap support **82** may support the lower end of the front panel **90**. The lower cap **80** may include the lower cap protrusion **83** protruding upward from the lower cap support **82**. When the front panel **90** is coupled to the door body **40**, the lower cap protrusion **83** may be inserted into the lower trim groove **131** of the lower trim **130**.

(118) Referring to FIG. **10**, the upper cap **60** of the door body **40** may be coupled to the upper portion of the door outer plate **41**. The upper cap **60** may form an upper surface of the door body **40**. The upper cap **60** may include an upper cap body **61**.

(119) The upper cap **60** may include a front upper wall **62** protruding forward from an upper portion of the upper cap body **61**. The front upper wall **62** may be elongated in the left and right direction.

(120) The upper cap **60** may include the trim receiving space **63** formed below the front upper wall **62**. The trim receiving space **63** may be elongated in the left and right direction.

(121) The upper cap **60** may include the trim opening **64** formed on the front side of the trim receiving space **63** to allow the upper trim **150** of the front panel **90** to be inserted into the trim receiving space **63**. The trim opening **64** may be elongated in the left and right direction.

(122) The upper cap **60** may include the fixer opening **65** formed on the upper side of the trim receiving space **63** to allow the fixer **79** to be inserted into the trim receiving space **63**. The fixer opening **65** may be formed at a position corresponding to the fixer coupling hole **153** of the upper trim **150**.

(123) The upper cap **60** may include a voice path **70** (refer to FIG. **15**) to guide external voice to the microphone module **170** mounted on the upper trim **150**. The voice path **70** will be described later.

(124) A process of coupling the front panel **90** to the door body **40** will be described with reference to FIGS. **8** to **10**. The lower cap protrusion **83** is inserted into the lower trim groove **131** by tilting the front panel **90**. In a state in which the lower cap protrusion **83** is inserted into the lower trim groove **131**, the front panel **90** becomes upright by rotating the front panel **90** around the lower cap protrusion **83**.

(125) While the front panel **90** rotates, the fixer coupler **152** of the upper trim **150** may be inserted into the trim receiving space **63** of the upper cap **60** through the trim opening **64** of the upper cap **60**. The fixer **79** is inserted into the trim receiving space **63** from top to bottom through the fixer opening **65** of the upper cap **60**. The fixer **79** may be coupled to the upper trim **150** in the trim receiving space **63**. The fixer **79** may be inserted into the fixer coupling hole **153** of the fixer coupler **152** of the upper trim **150**. The front panel **90** may be coupled to the door body **40** as the

fixer **79** is coupled to the upper trim **150**.

(126) Before standing the front panel **90** completely, the first connector **118** connected to the display substrate **190** of the front panel **90** may be coupled to the third connector **58** on the side of the door body **40**.

(127) In order to separate the front panel **90** from the door body **40**, the above sequence may be reversed. That is, the front panel **90** may be separated from the door body **40** while rotating the front panel **90** in the opposite direction after separating the fixer **79** from the upper trim **150** by pulling the fixer **79** up from the bottom.

(128) FIG. **11** is a view illustrating an upper fixer and a microphone module according to an embodiment of the disclosure. FIG. **12** is a bottom view of the upper fixer and the microphone module according to an embodiment of the disclosure, particularly illustrating a state in which the upper fixer and the microphone module are separated. FIG. **13** is a bottom view of the upper fixer and the microphone module according to an embodiment of the disclosure, particularly illustrating a state in which the microphone module is mounted to the upper fixer. FIG. **14** is an enlarged view of an upper portion of the upper cap according to an embodiment of the disclosure. FIG. **15** is a view illustrating a state in which a cap cover is separated from the upper cap according to an embodiment of the disclosure. FIG. **16** is a perspective cross-sectional view illustrating the microphone module attached to an upper trim and a voice path formed in the upper cap according to an embodiment of the disclosure. FIG. **17** is a view illustrating the microphone module attached to the upper trim and the voice path formed in the upper cap according to an embodiment of the disclosure, particularly a cross-sectional view taken along a Y-Z plane. FIG. **18** is a view illustrating the microphone module attached to the upper trim and the voice path formed in the upper cap according to an embodiment of the disclosure, particularly a cross-sectional view taken along a X-Y plane.

(129) The microphone module and the voice path according to an embodiment of the disclosure will be described with reference to FIGS. **11** to **18**.

(130) The microphone module **170** may include a microphone substrate **171** and a microphone **173** mounted on the microphone substrate **171**. The microphone substrate **171** may include a printed circuit board. The microphone substrate **171** may include a first surface **171a** and a second surface **171b** opposite to the first surface **171a**. The first surface **171a** may be referred to as a mounting surface **171a** on which the microphone **173** is mounted. The second surface **171b** may be referred to as an adhesive surface **171b**.

(131) The microphone **173** may include a microphone. The microphone **173** may be configured to convert an acoustic signal such as voice into an electrical signal. The microphone **173** may be a condenser type including a diaphragm.

(132) The microphone module **170** may include one or more microphones **173**. For example, the microphone module **170** may include a plurality of microphones **173**. Because the microphone module **170** includes the plurality of microphones **173**, noise may be effectively removed, and sound quality and recognition rate of the microphone module **170** may be improved.

(133) The microphone substrate **171** may include a substrate voice hole **172** formed at a position corresponding to the microphone **173**. The substrate voice hole **172** may be formed to penetrate the microphone substrate **171**. The substrate voice hole **172** may be formed in plurality at positions corresponding to the plurality of microphones **173**.

(134) The microphone module **170** may be mounted on the microphone module mounting member **154** of the upper trim **150**. The microphone module mounting member **154** may include a microphone module mounting space **155** formed on a lower surface of the microphone module mounting member **154**. The microphone module **170** may be mounted on the microphone module mounting member **154** to be received in the microphone module mounting space **155**. The microphone module **170** may be mounted on the microphone module mounting member **154** to allow the mounting surface **171a** of the microphone substrate **171** to face downward. That is, when

the microphone substrate **171** is mounted on the microphone module mounting member **154**, the microphone **173** may be disposed to face downward.

(135) When the upper trim **150** is received in the trim receiving space **63**, the microphone module **170** mounted on the upper trim **150** may also be received in the trim receiving space **63** together with the upper trim **150**.

(136) The upper trim **150** may include a trim voice hole **156** formed at a position corresponding to the substrate voice hole **172**. The trim voice hole **156** may be formed to penetrate the upper trim **150**. The trim voice hole **156** may be formed in the microphone module mounting member **154**. Voice received through the voice path **70** formed in the upper cap **60** may be received by the microphone **173** through the trim voice hole **156** and the substrate voice hole **172**.

(137) A sealing member **175** may be arranged between the microphone module mounting member **154** of the upper trim **150** and the microphone module **170**. The sealing member **175** may fix the microphone module **170** to the microphone module mounting member **154**. The sealing member **175** may prevent a gap from being generated between the lower surface of the microphone module mounting member **154** and the second surface **171b** of the microphone substrate **171**. That is, the sealing member **175** may allow the lower surface of the microphone module mounting member **154** and the second surface **171b** of the microphone substrate **171** to come into close contact with each other.

(138) For this, the sealing member **175** may be formed of an elastic rubber material or non-woven fabric. Adhesive may be applied on both sides of the sealing member **175**. The sealing member **175** may include a sealing member voice hole **176** to correspond to the trim voice hole **156** and the substrate voice hole **172**. As the lower surface of the microphone module mounting member **154** in close contact with the second surface **171b** of the microphone substrate **171** through the sealing member **176**, the voice passing through the trim sound hole **156** may be collected to the microphone **173** through the substrate voice hole **172** without leakage.

(139) The microphone module **170** may include a microphone connector **174** to be electrically connected to an external circuit. The microphone connector **174** may be provided on the mounting surface **171a** of the microphone substrate **171**. Accordingly, when the microphone module **170** is mounted on the microphone module mounting member **154** of the upper trim **150**, the microphone connector **174** may be positioned to face the lower side of the microphone substrate **171**. Therefore, in a state in which the microphone module **170** is mounted on the microphone module mounting member **154** of the upper trim **150**, other connectors may be easily coupled to the microphone connector **174**.

(140) The upper trim **150** may include a connector receiving member **160** to easily couple other connectors to the microphone connector **174**. The connector receiving member **160** may be formed by cutting a portion of the upper trim **150**. The connector receiving member **160** may be formed by cutting the upper trim body **151** of the upper trim **150**. A space, in which at least a portion of the microphone connector **174** is disposed, may be formed in the connector receiving member **160**. A space, in which at least a portion of other connector coupled to the microphone connector **174** is disposed, may be formed in the connector receiving member **160**. For example, at least a portion of the second connector **120** (refer to FIG. 6) coupled to the microphone connector **174** may be disposed in the connector receiving member **160**.

(141) Referring FIGS. 14 to 18, the upper cap **60** may include the voice path **70** for receiving voice. The voice path **70** may receive external voice and guide the received external voice to the trim voice hole **156** of the upper trim **150**. The voice guided to the trim voice hole **156** may be collected to the microphone **173** through the substrate voice hole **172**.

(142) The voice path **70** may include a voice guide hole **73** formed in the front upper wall **62** of the upper cap **60**. The voice guide hole **73** may be formed to vertically penetrate the upper front wall **62**.

(143) The upper cap **60** may include a cap protrusion **66** protruding upward from the front upper

wall **62** to form at least a portion of the voice path **70**. The cap protrusion **66** may be formed to protrude upward from a periphery of the voice guide hole **73** of the front upper wall **62**. The cap protrusion **66** may be integrally formed with the upper cap **60**. That is, the cap protrusion **66** may be integrally injection molded with the upper cap **60**. Alternatively, the cap protrusion **66** may be formed separately from the upper cap **60** and then coupled to the upper surface of the upper cap **60**. (144) The cap protrusion **66** may be formed to be gently inclined toward the center of the cap protrusion **66** from the lower side to the upper side. As shown in FIG. **14**, the cap protrusion **66** may be provided in plurality to correspond to the plurality of microphones **173**. The plurality of cap protrusions **66** may be connected by a protrusion connector **68**.

(145) The voice path **70** may include a voice guide space **72** formed inside the cap protrusion **66**. The voice guide space **72** may be a space surrounded by the cap protrusion **66**. The voice guide hole **73** may be connected to the lower side of the voice guide space **72**. That is, the lower side of the voice guide space **72** may be open.

(146) An upper side of the voice guide space **72** may be open. That is, a protrusion upper opening **67** may be formed on the upper side of the voice guide space **72**. The protrusion upper opening **67** may be covered by a cap cover **75** to be described later.

(147) However, according to embodiments, the cap protrusion **66** may be formed to allow the upper side of the voice guide space **72** not to be opened.

(148) The cap protrusion **66** may include a cap front protrusion **66a**, a cap rear protrusion **66b**, a cap left protrusion **66c**, and a cap right protrusion **66d**. That is, the cap protrusion **66** may have a substantially rectangular shape when viewed from above. However, the shape of the cap protrusion **66** is not limited thereto, and other shapes such as a triangle and a circle are also possible. That is, it is sufficient that the cap protrusion **66** forms a closed space on the X-Y plane to form the voice guide space **72**.

(149) The voice path **70** may include a voice input hole **71** formed in the cap protrusion **66**. The voice input hole **71** may be formed in at least one of the cap front protrusion **66a**, the cap rear protrusion **66b**, the cap left protrusion **66c**, and the cap right protrusion **66d**. For example, the voice input hole **71** may be formed in the cap front protrusion **66a**. Because the voice input hole **71** is formed in the cap front protrusion **66a**, voice generated from the front of the refrigerator may be efficiently received.

(150) The voice input hole **71** may be formed to penetrate the cap protrusion **66**. Accordingly, the voice input hole **71** may be connected to the voice guide space **72**.

(151) The voice input hole **71** may be formed to be spaced apart from the lower surface **62a** of the front upper wall **62** of the upper cap **60** by a predetermined distance D (refer to FIG. **17**). That is, the voice input hole **71** may be separated from the trim receiving space **63**. The voice input hole **71** may be spaced apart from the upper end of the front panel **90**.

(152) Instead of the front panel **90**, a portion of the front upper wall **62** may be positioned right under the voice input hole **71**. The lower side of the voice input hole **71** may not be opened. Therefore, when viewed from the front of the refrigerator, the lower surface **62a** of the front upper wall **62** may be formed in a straight line without a cut in the left and right direction. Because the lower surface **62a** of the front upper wall **62** is formed in a straight line without a cut in the left and right direction when viewed from the front of the refrigerator, aesthetics may be improved.

(153) The upper cap **60** may include the cap cover **75** provided to cover the protrusion upper opening **67**. Because the cap cover **75** covers the protrusion upper opening **67**, it is possible to prevent a state in which moisture or other foreign substances penetrate into the voice guide space **72** through the protrusion upper opening **67**.

(154) The cap cover **75** may be coupled to the upper cap **60**. For example, the cap cover **75** may be inserted into the voice guide space **72** through the protrusion upper opening **67**. Alternatively, without being inserted into the voice guide space **72**, the cap cover **75** may be provided to cover the protrusion upper opening **67** on the outside of the protrusion upper opening **67**. The cap cover **75**

may be provided to cover other areas of the upper cap **60** as well as the protrusion upper opening **67**. Furthermore, the cap cover **75** may be provided to cover the entire upper surface of the upper cap **60**.

(155) The cap cover **75** may include a cap cover body **76** provided to have a shape corresponding to the protrusion upper opening **67** so as to cover the protrusion upper opening **67**. A lower surface of the cap cover body **76** may form at least a part of the voice guide space **72**. For example, the lower surface of the cap cover body **76** may form the upper side of the voice guide space **72**.

(156) The cap cover **75** may include a cap cover leg **77** extending downward from the cap cover body **76**. One or more cap cover legs **77** may be formed. For example, the cap cover leg **77** may be formed in plurality. The plurality of cap cover legs **77** may be formed to be spaced apart from each other in the left and right direction (X-axis direction). Voice may be guided between the plurality of cap cover legs **77** spaced apart from each other.

(157) A cover hooking member **78** may be formed at a lower end of the cap cover leg **77**. The cover hooking member **78** may be provided to hook to the lower surface **62a** of the front upper wall **62** of the upper cap **60**.

(158) When the cap cover **75** is inserted into the voice guide space **70** through the protrusion upper opening **67**, the cap cover leg **77** may be elastically deformed in a retracting direction and at the same time, the cover hooking member **78** may be moved to under the front upper wall **62**. When the cap cover **75** is inserted to allow the cover hooking member **78** to be positioned under the front upper wall **62**, the cap cover leg **77** may return to its original shape in the direction in which the cap cover leg **77** opens and at the same time, the cover hooking member **78** may hook to the lower surface **62a** of the front upper wall **62**. Accordingly, the cap cover **75** may be coupled to the upper cap **60**.

(159) The upper trim **150** may include a cap cover receiving member **158** to receive at least a portion of the cap cover leg **77**. The cap cover receiving member **158** may be concavely formed on an upper surface of the microphone mounting member **154**.

(160) The upper trim **150** may include a trim protrusion **159** provided to be coupled to the cap cover **75**. The trim protrusion **159** may protrude upward from the upper trim **150**. When the upper trim **150** is inserted into the trim receiving space **63** of the upper cap **60** and the cap cover **75** is coupled to the upper cap **60**, the trim protrusion **159** may be coupled to the cap cover **75**.

(161) The trim protrusion **159** may be coupled to the cover leg **77** of the cap cover **75**. The trim protrusion **159** may support one surface of the cover leg **77** to prevent the upper trim **150**, which is inserted into the trim receiving space **63**, from being separated from the trim receiving space **63**.

(162) With the above structure, the cap cover **75** may fix the front panel **90** to the door body **40** together with the fixer **79**.

(163) As described above, the voice path **70** formed in the upper cap **60** may include the voice input hole **71**, the voice guide space **72**, and the voice guide hole **73**. External voice may pass through the voice path **70** of the upper cap **60** and be transferred to the microphone module **170** disposed in the trim receiving space **63** of the upper cap **60**.

(164) The voice path **70** may be provided in plurality to correspond to the number of microphones **173** included in the microphone module **170**. The plurality of voice paths **70** may be provided to be separated from each other. A downstream end of the voice path **70** may be arranged to be perpendicular to the microphone **173** of the microphone module **170** to allow voice to be efficiently transmitted to the microphone **173**. That is, the trim voice hole **156** of the upper trim **150** and the substrate voice hole **172** of the microphone module **170** may be positioned below the voice guide space **72** and the voice guide hole **73** in the vertical direction.

(165) FIG. **19** is other view illustrating the microphone module attached to the upper trim and the voice path formed in the upper cap according to an embodiment of the disclosure, particularly a cross-sectional view taken along the X-Y plane.

(166) Referring to FIG. **19**, the downstream end of the voice path **70** and the microphone **173** of the

microphone module **170** may be misaligned so as not to be perpendicular to each other. That is, the trim voice hole **156** of the upper trim **150** and the microphone module **170** may be arranged vertically under the voice guide space **72** and the voice guide hole **73** to be horizontally spaced apart by a predetermined distance **L** from the voice guide space **72** and the voice guide hole **73**.

(167) The cap cover receiving member **158a** may be formed to extend in the left and right direction (X-axis direction) to connect the voice guide hole **73** and the trim voice hole **156**.

(168) As mentioned above, as the downstream end of the voice path **70** and the microphone **173** of the microphone module **170** are misaligned so as not to be perpendicular to each other, it is possible to prevent a state in which moisture or foreign substances penetrate into the microphone **173** through the voice path **70**.

(169) FIG. **20** is a perspective cross-sectional view illustrating a microphone module attached to an upper trim and a voice path formed between a stepped member of an upper cap and a front panel according to an embodiment of the disclosure.

(170) The same reference numerals are assigned to components the same as those of the above-described embodiment, and descriptions thereof may be omitted.

(171) The upper cap **60** may include the front upper wall **62** protruding forward from the upper portion of the upper cap body **61**. The upper cap **60** may include the trim receiving space **63** formed below the front upper wall **62**.

(172) The upper cap **60** may include a voice path **270** provided to guide external voice to the microphone module **170** mounted on the upper trim **150**. The voice path **270** may receive external voice and guide the voice to the trim voice hole **156** of the upper trim **150**. The voice guided to the trim voice hole **156** may be collected to the microphone **173** through the substrate voice hole **172**.

(173) The upper cap **60** may include a stepped member **266** formed by bending upward from the front upper wall **62** to form at least a portion of the voice path **270**. A slit **S** may be formed between a front end of the stepped member **266** and the front panel **90**. The slit **S** may have a height sufficient to allow voice to pass through. The slit **S** may be elongated in the left and right direction. The voice path **270** may include the slit **S** formed between the front end of the stepped member **266** and the front panel **90**.

(174) As described above, the voice path **270** may be implemented in the upper cap **60** with a relatively simple configuration.

(175) The above mentioned-disclosure is not applied only to the above-described refrigerator and the door of the refrigerator, but may also be applicable to a home appliance including a main body including a cavity, which is an internal space for performing a specific function, and a door configured to open and close the cavity, and the door of the home appliance.

(176) For example, the above mentioned-disclosure may be applied to a cooking appliance including a cooking chamber, a dishwasher including a washing chamber, a dryer including a drying chamber, a clothes treating device including a clothes treating chamber, and an air conditioner including a heat exchange chamber in which a heat exchanger is disposed.

(177) As is apparent from the above description, a refrigerator may include a door body rotatably coupled to a main body so as to open and close a storage compartment, and a front panel detachably mounted on the door body and including a microphone module arranged on a rear surface of the front panel. However, a microphone hole provided to transmit voice to the microphone module may not be provided on the front panel.

(178) Therefore, aesthetics may be improved because the microphone hole is omitted. Further, cost may be reduced because a process for forming the microphone is not required.

(179) Although a few embodiments of the disclosure have been shown and described, it would be appreciated by those skilled in the art that changes may be made in these embodiments without departing from the principles and spirit of the disclosure, the scope of which is defined in the claims and their equivalents.

Claims

1. A refrigerator comprising: a main body having a storage compartment; a door body rotatably coupled to the main body to open and close the storage compartment, the door body including: a door outer plate forming a front surface of the door body, and an upper cap coupled to an upper portion of the door outer plate, the upper cap having a fixer opening on an upper side of the upper cap, a trim opening on a front side of the upper cap, and a trim receiving space; a front panel including an upper trim, the upper trim including an upper trim body and a microphone module mounting member extending rearward from the upper trim body, and the microphone module mounting member including a microphone module mounting space on a lower surface of the microphone module mounting member; a fixer; and a microphone module mounted on the microphone module mounting member to be received in the microphone module mounting space to thereby be coupled to the upper trim, wherein the door body, the front panel, and the fixer are configured so that the upper trim is insertable into the trim receiving space through the trim opening, and, with the upper trim inserted into the trim receiving space through the trim opening, the fixer is insertable into the trim receiving space through the fixer opening to be coupled to the upper trim, to detachably couple the front panel to a front side of the door body with the microphone module disposed in the trim receiving space to receive voice through a voice path formed in the upper cap.
2. The refrigerator of claim 1, wherein the upper cap includes: a cap body, a front upper wall on a front upper side of the cap body, so that the trim receiving space is below the front upper wall, and a voice guide hole penetrating the front upper wall and forming a part of the voice path.
3. The refrigerator of claim 2, wherein the upper cap includes a cap protrusion protruding upward from a periphery of the voice guide hole, and a voice guide space surrounded by the cap protrusion and forming a part of the voice guide path.
4. The refrigerator of claim 3, wherein the cap protrusion includes a cap front protrusion in front of the voice guide space, and a voice input hole is formed in the cap front protrusion and forms a part of the voice path.
5. The refrigerator of claim 4, wherein the voice input hole is spaced apart from a lower surface of the front upper wall by a predetermined height.
6. The refrigerator of claim 3, wherein the upper cap includes a protrusion upper opening on an upper side of the voice guide space, and the upper cap includes a cap cover that is coupleable to the upper cap to cover the protrusion upper opening.
7. The refrigerator of claim 6, wherein the cap cover includes: a cover body configured to cover the protrusion upper opening with a lower surface of the cover body forming at least a part of the voice path, and a cover leg extending downward from the lower surface of the cover body and including a cover hooking member configured to hook to a lower surface of the front upper wall.
8. The refrigerator of claim 6, wherein the upper trim includes a trim protrusion protruding upward and configured to be coupled to the cap cover in a state in which the upper trim is inserted into the trim receiving space and the cap cover is coupled to the upper cap.
9. The refrigerator of claim 1, wherein the microphone module includes a microphone substrate, and a microphone on a mounting surface of the microphone substrate, the microphone module is mounted in the microphone module mounting space with the mounting surface of the microphone substrate facing downward, and the microphone substrate includes a substrate voice hole penetrating the microphone substrate at a position corresponding to the microphone.
10. The refrigerator of claim 9, wherein the upper trim includes a trim voice hole at a position corresponding to the substrate voice hole.
11. The refrigerator of claim 10, further comprising: a sealing member between the microphone module mounting member and the microphone module, wherein the sealing member includes a

sealing member voice hole at a position corresponding to the trim voice hole and the substrate voice hole.

12. The refrigerator of claim 9, wherein the microphone module includes a connector electrically connected to an external circuit, and the connector is on the mounting surface of the microphone substrate and positioned below the microphone substrate.

13. The refrigerator of claim 12, wherein the upper trim includes a connector receiving member configured to receive the connector or other connector connected to the connector.

14. The refrigerator of claim 1, wherein the microphone module includes a microphone substrate, and a plurality of microphones spaced apart from each other on a mounting surface of the microphone substrate, and the voice path includes a plurality of voice paths separated from each other to correspond to the plurality of microphones.
